Combination circuits

antocciphysics | TEACHER KEY | CPA

Learning target: Reduce a circuit by identifying genuine series and parallel groups.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 6; 3 course days later.*
V=IR=15 V.

02. Explain a worked example (worked)
Series resistor drop 6 V; parallel section 6 V; each 4 ohm branch 1.5 A.

03. Look and explain (model)
One resistor before a split into two 6 Ω branches, then return to supply.

04. Build the solution (practice)
5 Ω total; 2 A.

05. Use the idea (reason)
Electrical connections and shared nodes, not page position.

06. Choose a method (mixed) *Review: Unit 6, Day 15; 12 course days later.*
3000 m; range is half the full path.

07. Exit check (exit)
1 A each.

08. STRETCH: Apply it another way (stretch)
Parallel 4 ohms; total 6 ohms; current 2 A.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 6; 3 course days later.*
R=30 Ω; V/I is constant at these points under the measured conditions.

02. Explain a worked example (worked)
Series resistor drop 6 V; parallel section 6 V; each 4 ohm branch 1.5 A.

03. Represent and interpret (model)
One resistor before a split into two 6 Ω branches, then return to supply.

04. Solve and justify (practice)
$R_{parallel}$=3 Ω; total 5 Ω; I=2 A; series drop 4 V; branch voltage 6 V; each branch 1 A.

05. Analyze the reasoning (reason)
Electrical connections and shared nodes, not page position.

06. Choose a method (mixed) *Review: Unit 6, Day 15; 12 course days later.*
At the same wave speed, equal range gives equal out-and-back time. Doppler shift depends on radial velocity, so motion can differ.

07. Exit check (exit)
Parallel equivalent 2 Ω; total 4 Ω; current 2.5 A.

08. STRETCH: Apply it another way (stretch)
Parallel 2, total 5 ohms; total 3 A; parallel voltage 6 V; branches 1 and 2 A.